

Workshop 1: Vector Equations

Name:

BUID:

1. Compute the following linear combination.

$$\begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - 3 \begin{bmatrix} 7 \\ 5 \\ 0 \end{bmatrix}$$

2. Determine the linear system whose solution set is identical to this vector equation.

$$x_1 \begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - x_3 \begin{bmatrix} 7 \\ 5 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix}$$

3. Draw the following vectors.

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

4. Can $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ be written as a linear combination of $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} -4 \\ -6 \end{bmatrix}$? Justify your answer.

5. Determine if $\begin{bmatrix} 2 \\ 3 \end{bmatrix} \in \text{span} \left\{ \begin{bmatrix} -4 \\ 3 \end{bmatrix}, \begin{bmatrix} -9 \\ 7 \end{bmatrix} \right\}$. If so, write it as a linear combination.

6. For what values of h and k is $\begin{bmatrix} 3 \\ k \end{bmatrix} \notin \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} h \\ -5 \end{bmatrix} \right\}$?

Write your solutions below and on the back on this page.